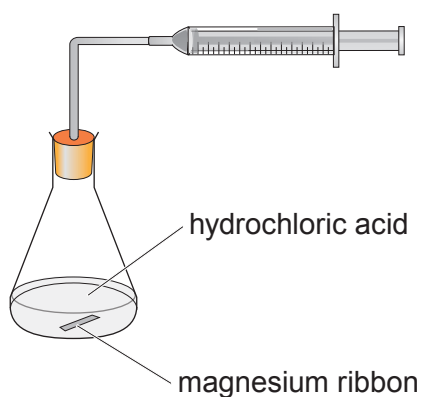
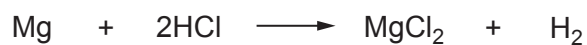


8. A group of students investigated the rate of the reaction between magnesium and dilute hydrochloric acid.

The equation for the reaction is as follows.



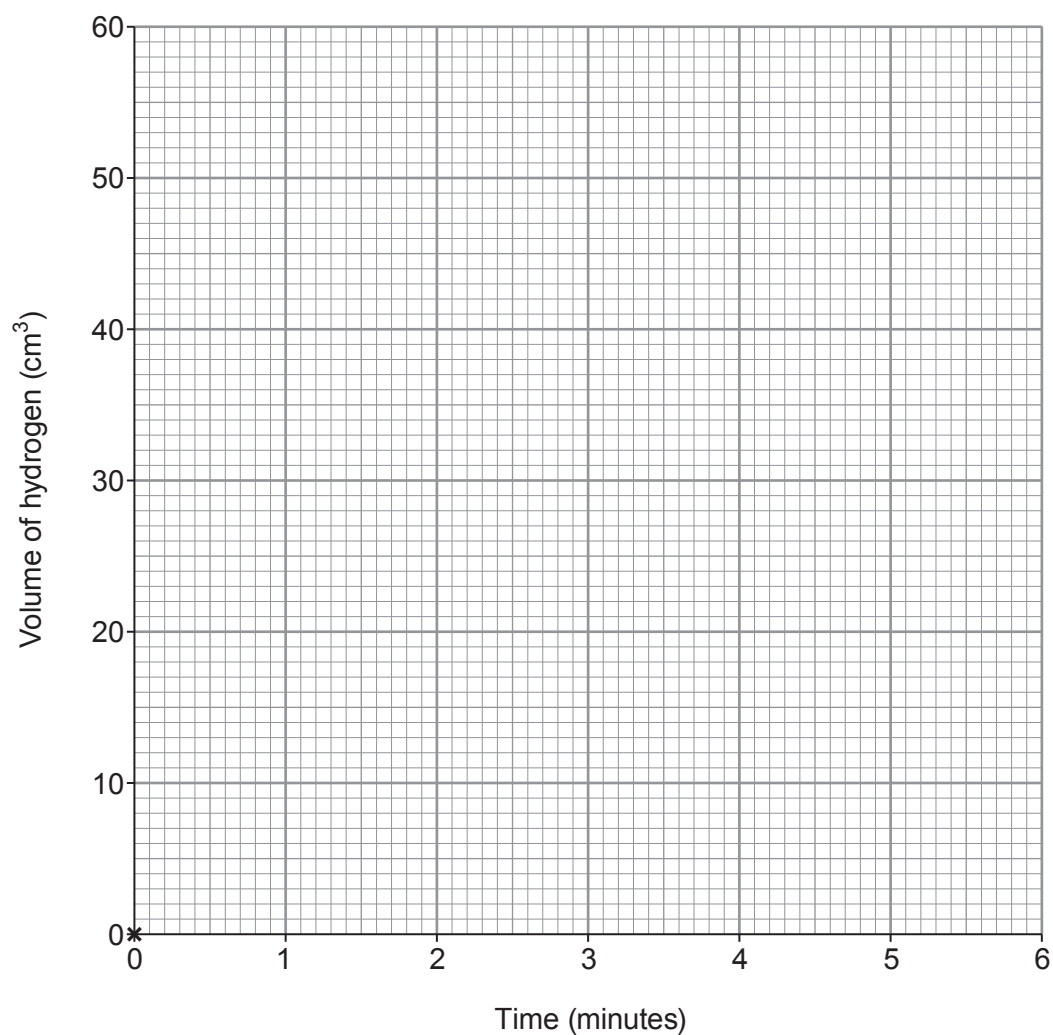
They carried out the reaction at 30 °C. The hydrogen gas was collected in a gas syringe and the volume recorded every minute for 6 minutes.

The results are shown in the table. The value at 1 minute has been left out.

Time (minutes)	0	1	2	3	4	5	6
Volume of hydrogen (cm ³)	0		29	39	46	50	50



- (a) (i) Plot the volume of hydrogen produced against time on the grid. The first point has been plotted for you. Draw a suitable line. [3]



- (ii) I. Use your graph to estimate the volume of hydrogen that would have been produced after 1 minute. [1]

..... cm³

- II. Calculate the mean rate of the reaction over the **first** minute. Give your answer in **cm³/s**. [2]

Use the formula

$$\text{mean rate} = \frac{\text{volume of hydrogen (cm}^3\text{)}}{\text{time (s)}}$$

Mean rate = cm³/s

- (b) There is no catalyst for this reaction.

Give **two** ways the students could increase the rate of this reaction. [2]

.....
.....

- (c) The students calculated that if they used 0.5g of magnesium in this reaction, they would make 2.0g of magnesium chloride. However, when they used 0.5g of magnesium only 1.7g of magnesium chloride was made.

Calculate the percentage yield for this reaction. [2]

Percentage yield = %

